

1. (15%) Convert the expressions (a) $(A+B)C-D/E$ and (b) $A/B * C + DE * A * C$ into prefix and postfix forms.
2. (15%) Given the recursive syntax for the Towers of Hanoi as follows:

Towers_Of_Hanoi (N, A, B, C)
Begin
 If $N > 0$ Then
 Towers_Of_Hanoi (N-1, A, C, B)
 Move disk N from A to B
 Towers_Of_Hanoi (N-1, C, B, A)
 End If
End

Parameter Explanation: N represents the number of disks to be moved, A is the source, B is the destination, and C is the auxiliary pole used to move N disks from A to B. Please convert the above recursive code into a non-recursive code.
3. (10%) Suppose there is an array of size n, and the data in the array is randomly distributed. Without changing the current positions of data in the array, is there a way to accelerate the search for specific data, and how can this be achieved? Please provide a brief explanation.
4. (10%) Let $A[1..j, 1..k]$ be a two-dimensional array. Each element of A requires two bytes. Suppose $A[2, 2]$ is stored at 898, and $A[1, 3]$ is stored at 910. Assuming that memory is addressed in bytes, and the array is stored sequentially in memory. This array is column-major order.
 - (a) What is the value of k?
 - (b) What is the address of $A[5, 7]$?
5. (10%) Analyze the time and space complexity of the following code segment:

(a)

```
for i:=1 to m do
  for j:=1 to p do
    begin
      c[i, j] = 0 ;
      for k:=1 to n do
        c[i, j] = c[i, j] + a[i, k] + b[k, j] ;
      end ;
    end ;
```

(b)

```
function s(var a:anArray ; n:integer):real
begin
  if n <= 0 then s:=0
  else s:=s(a, n-1)+a[n]
end
```

6. (15%) A stack is a last-in and first-out array list.

(a) Design a data structure to represent a stack.

(b) Design an algorithm to push elements onto your stack.

(c) Design an algorithm to pop elements from your stack.

7. (15%) The function defined as follows is a standard device for determining recursive executions on a computer.

$A(0, n) = n + 1$ for $n \geq 0$

$A(m, 0) = A(m - 1, 1)$ for $m > 0$ and $n = 0$

$A(m, n) = A(m - 1, A(m, n - 1))$ for $m > 0$ and $n > 0$

Please calculate the following values: $A(0, 0)$, $A(1, 8)$, $A(2, 0)$.

8. (10%) Consider the following code segment:

Program main() ;

int x, y ;

procedure SUB(int a, b, c,d) ;

begin

b:= a+d ;

a:= c ;

d:=d+1 ;

return

end ;

begin

x:= 1 ; y:= 2 ;

SUB(x, y, x+y, y) ;

End

What are the final values of the parameters, x and y, when processed

(a) by value and (b) by address?